Order Hymenoptera

Family Formicidae

Ants

Occurrence 12,000 spp. worldwide; in virtually all regions except in Antarctica and on a few oceanic islands

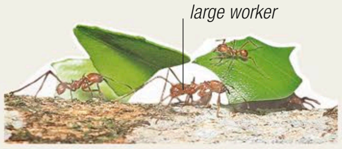
Habitat Most terrestrial habitats

Ants are found almost everywhere on Earth and have an immense impact on terrestrial ecosystems. In many, ants move more earth than earthworms and are vital in nutrient recycling and seed dispersal. Like termites and some bees and wasps, ants are highly social and live in colonies ranging from a handful of individuals sharing a tiny shelter to tens or hundreds of millions inhabiting structures reaching 20 ft (6 m) below ground. Two or more generations of ants overlap at any one time, and adults take care of the young. There are different classes (or castes) of ants within a colony, each with a specific task to perform: some are workers, who also defend the nest; others are dedicated to reproduction. Caste is determined by diet: a rich diet producing reproductives, a poor one producing sterile workers. Mating may take place on the wing or on the ground, after which the males die and the females lose their wings. All ants have glands that secrete pheromones—chemical messages used for communication between colony members as well as for trail making and defense. Numerous species of insects and plants have evolved symbiotic and often highly complex relationships with ants. In some cases, plants provide ants with food or homes; in return, the ants may protect the plants by disturbing potentially damaging herbivores, or stripping leaves from encroaching vines. The ant family is divided into 10 distinct subfamilies, of which the biggest by far are the Myrmicinae and the Formicinae. Some myrmicine ants have stings, while formicines defend themselves by spraying formic acid. The infamous driver and army ants of tropical regions belong to the subfamily Dorylinae. Their colonies move in massive columns, up to several million strong, often raiding termite or ant nests.

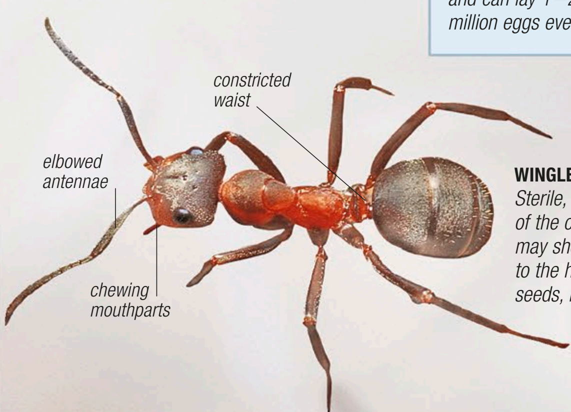
EATING HABITS

Some ants are herbivorous seed gatherers or fungus eaters. Other species are carnivorous or omnivorous, while some rely exclusively on the honeydew produced by sap-sucking bugs.

WORKERS WITH PREY
Worker wood ants, Formica rufa, *may cooperate in dragging a large item of prey back to the nest, where it will be cut up.*



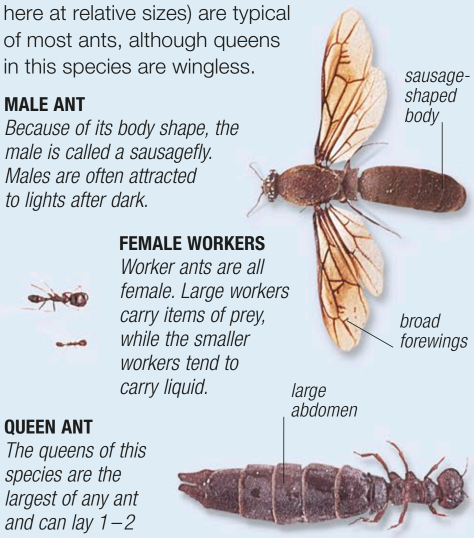
LEAFCUTTERS
Leafcutter ants, Atta *spp., chew pieces of leaves into pulp, which they use to fertilize fungus beds—their sole source of food.*



WINGLESS FEMALE
Sterile, wingless females do the work of the colony, and in some species may show amazing adaptations to the head and jaws for crushing seeds, blocking nest entrances, dismembering enemies, and other purposes.

DIFFERENT CASTES

A colony will contain 3 castes of ants: workers, which are always wingless sterile females, queens, and males (which are usually winged and form mating swarms with queens at certain times of the year). The physical differences between the castes of the African driver ant *Dorylus nigricans* (shown here at relative sizes) are typical of most ants, although queens in this species are wingless.



MALE ANT
Because of its body shape, the male is called a sausagefly. Males are often attracted to lights after dark.

FEMALE WORKERS
Worker ants are all female. Large workers carry items of prey, while the smaller workers tend to carry liquid.

QUEEN ANT
The queens of this species are the largest of any ant and can lay 1–2 million eggs every month.

ON THE DEFENSIVE
This worker wood ant, Formica rufa, *has taken a defensive posture with its jaws wide open and abdomen curled under and forward. In this position, it will be able to spray formic acid toward an attacker.*